
title: Ethereum QuickStart

description: Get started with Ethereum RPC on Quicknode. Follow this QuickStart to get up and running with your own Quicknode endpoint and making your first API request to the Ethereum blockchain.

The quickest way to start building on Ethereum with Quicknode is by sending a JSON-RPC request to your endpoint. In this quickstart, you'll create an endpoint, copy its provider URL, and make your first request. Code samples are available in ****cURL**** as well as popular SDKs and programming languages.

Get Your Ethereum Endpoint

> For the complete documentation index, see [llms.txt](/docs/llms.txt)

1

Create a Quicknode account

Sign up [here](https://www.quicknode.com/signup?utm_source=internal&utm_campaign=docs&utm_content=quickstart-Ethereum) if you haven't already.

2

Go to your dashboard

Go to your Quicknode dashboard, then select [Endpoints] (https://dashboard.quicknode.com/endpoints) from the left sidebar.

3

Create a new endpoint

Click ****Create Endpoint**** in the top-right corner. Select ****Ethereum**** as the chain, click ****Continue****, then select your preferred network.

4

Copy your provider URLs

Keep the HTTP URL handy. You'll use it in your requests below.

Learn More about the Dashboard

For a detailed walkthrough of the Quicknode dashboard, check out our [guide] (https://www.quicknode.com/guides/quicknode-products/how-to-use-the-quicknode-dashboard)

Send Your First Request

Your endpoint is ready. Now, let's make your first call to the Ethereum blockchain. We'll use the ``eth_blockNumber`` method, which returns the latest block number. ****Select your preferred language or SDK and follow the steps below to send your first request****.

****cURL****

1

Check cURL installation

Most *nix based systems have cURL support out of the box. Open your terminal and check the cURL version by running the command below:

```
```bash
curl --version
```
```

2

Send a JSON-RPC request

In your terminal, copy and paste the following cURL command to retrieve the latest block number:

```
```bash
curl -X POST YOUR_QUICKNODE_ENDPOINT_URL/ \
-H "Content-Type: application/json" \
-d '{"jsonrpc":"2.0","method":"eth_blockNumber","params":[],"id":1}'
```
```

3

Sample Response

```
```json
{
 "jsonrpc": "2.0",
 "id": 1,
 "result": "0x1234567"
}
```
```

****SDK****

1

Set up your project

First, verify Node.js is installed with `node --version`. If not installed, download it from <https://nodejs.org>. Then create a new directory and initialize a Node.js project:

```
```bash
```

```
mkdir ethereum-api-quickstart
```

```
cd ethereum-api-quickstart
```

```
npm init -y
```

2

```
Install the SDK
```

```
Install the SDK package:
```

```
```bash
npm install @quicknode/sdk
```
```

3

```
Create your main file
```

```
Create an index.js file for your code:
```

```
```bash
touch index.js
```
```

4

```
Add this code to your index.js file
```

```
Copy and paste this code into your index.js file:
```

```
```javascript
const { Core } = require('@quicknode/sdk');
```

```
const main = async () => {
```

```
  const core = new Core({
    endpointUrl: 'YOUR_QUICKNODE_ENDPOINT_URL/',
  });
```

```
  const blockNumber = await core.client.request({
    method: 'eth_blockNumber',
  });
```

```
  console.log('Block number:', parseInt(blockNumber, 16));
```

```
};
```

```
main();  
````
```

5

```
Run your project
```

Execute your script to see the latest block number:

```
```bash  
node index.js  
```
```

```
Python
```

1

```
Set up your project
```

Create a new directory for your Python project:

```
```bash  
mkdir ethereum-python-quickstart  
  
cd ethereum-python-quickstart  
```
```

2

```
Create and activate a virtual environment
```

Create a virtual environment to manage dependencies:

```
```bash  
python3 -m venv venv  
  
source venv/bin/activate  
```
```

3

```
Install requests
```

Install the requests library for making HTTP requests:

```
```bash  
pip install requests  
```
```

4

```
Create a Python script (app.py)
```

Create a Python file with the following code:

```

```python
import requests

import json

# Your Quicknode endpoint URL
endpoint_url = "YOUR_QUICKNODE_ENDPOINT_URL/"

# JSON-RPC request payload
payload = {
    "jsonrpc": "2.0",
    "id": 1,
    "method": "eth_blockNumber",
    "params": []
}

# Make the request
response = requests.post(endpoint_url, json=payload)
result = response.json()

print("Latest block number (hex):", result["result"])
print("Latest block number (decimal):", int(result["result"], 16))
```

5

Run the script

Execute your Python script:

```bash
python app.py
```

JavaScript

1

Set up your project

```

Create a new directory and initialize a Node.js project:

```
```bash
mkdir ethereum-js-quickstart

cd ethereum-js-quickstart

npm init -y
```
```

2

### Create a JavaScript file (app.js)

Create an app.js file with the following code (using native fetch available in Node.js 18+):

```
```javascript
const url = 'YOUR_QUICKNODE_ENDPOINT_URL/';

const options = {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    jsonrpc: '2.0',
    method: 'eth_blockNumber',
    params: [],
    id: 1
  })
};

async function main() {
  try {
    const response = await fetch(url, options);
    const data = await response.json();
    console.log('Block Number (hex):', data.result);
    console.log('Block Number (decimal):', parseInt(data.result, 16));
  } catch (error) {
    console.error('Error:', error);
  }
}
```

```
}  
}
```

```
main();  
```
```

3

### Run the script

Execute your JavaScript script:

```
```bash  
node app.js  
```
```

**\*\*Ruby\*\***

1

### Set up your project

Create a new directory for your Ruby project:

```
```bash  
mkdir ethereum-ruby-quickstart  
  
cd ethereum-ruby-quickstart  
```
```

2

### Check Ruby installation

Verify Ruby is installed on your system. If not, install it from <https://ruby-lang.org>:

```
```bash  
ruby --version  
```
```

3

### Create a Ruby script (main.rb)

Create a main.rb file with the following code:

```
```javascript  
require "uri"  
  
require "json"  
  
require "net/http"
```

```

url = URI("YOUR_QUICKNODE_ENDPOINT_URL/")

https = Net::HTTP.new(url.host, url.port)
https.use_ssl = true

request = Net::HTTP::Post.new(url)
request["Content-Type"] = "application/json"
request.body = JSON.dump({
  "method": "eth_blockNumber",
  "params": [],
  "id": 1,
  "jsonrpc": "2.0"
})

response = https.request(request)
result = JSON.parse(response.read_body)
hex_result = result["result"]
decimal_result = hex_result.to_i(16)
puts "Block number (hex): #{hex_result}"
puts "Block number (decimal): #{decimal_result}"
```

4

Run the script

Execute your Ruby script to see the latest block number:

```bash
ruby main.rb
```

Go

1

Set up your project

```

Create a new directory for your Go project:

```
```bash
mkdir ethereum-go-quickstart

cd ethereum-go-quickstart
```
```

2

### Check Go installation

Verify Go is installed on your system. If not, install it from <https://golang.org>:

```
```bash
go version
```
```

3

### Create a Go script (main.go)

Create a main.go file with the following code:

```
```go
package main

import (
    "encoding/json"
    "fmt"
    "strings"
    "net/http"
    "io/ioutil"
    "strconv"
)

type Response struct {
    Result string `json:"result"`
}

func main() {
```

```

url := "YOUR_QUICKNODE_ENDPOINT_URL/"

method := "POST"

payload := strings.NewReader(`{"method":"eth_blockNumber","params":
[],"id":1,"jsonrpc":"2.0"}`)

client := &http.Client{}

req, err := http.NewRequest(method, url, payload)

if err != nil {
    fmt.Println(err)
    return
}

req.Header.Add("Content-Type", "application/json")

res, err := client.Do(req)

if err != nil {
    fmt.Println(err)
    return
}

defer res.Body.Close()

body, err := ioutil.ReadAll(res.Body)

if err != nil {
    fmt.Println(err)
    return
}

var response Response

err = json.Unmarshal(body, &response)

```

```

if err != nil {
    fmt.Println("Error parsing JSON:", err)
    return
}

// Convert hex to decimal
blockNumber, err := strconv.ParseInt(response.Result, 0, 64)
if err != nil {
    fmt.Println("Error converting hex to decimal:", err)
    return
}

fmt.Printf("Block number (hex): %s\n", response.Result)
fmt.Printf("Block number (decimal): %d\n", blockNumber)
}

```

4

Run the script

Execute your Go script to see the latest block number:

```

```bash
go run main.go
```

```

****Ethers.js****

1

Set up your project

Create a new directory and initialize a Node.js project (assumes Node.js is already installed):

```

```bash
mkdir ethereum-ethers-quickstart
cd ethereum-ethers-quickstart
npm init -y

```

```
npm pkg set type=module
```

2

```
Install ethers.js
```

Install the ethers.js library for Ethereum interaction:

```
```bash
npm install ethers
```
```

3

```
Create your main file (index.js)
```

Create an index.js file with the following code:

```
```javascript
import { ethers } from "ethers";

(async () => {
  const provider = new ethers.JsonRpcProvider("YOUR_QUICKNODE_ENDPOINT_URL/");
  const blockNum = await provider.getBlockNumber();
  console.log('Current block number (decimal):', blockNum);
  console.log('Block number (hex):', '0x' + blockNum.toString(16));
})();
```
```

4

```
Run your project
```

Execute your script to see the latest block number:

```
```bash
node index.js
```
```

```
Web3.py
```

1

```
Set up your project
```

Create a new directory for your web3.py project:

```
```bash
mkdir ethereum-web3py-quickstart
```

```
cd ethereum-web3py-quickstart
```

```
```
```

2

```
Install web3.py (if not already installed)
```

Check if web3.py is installed with `\`pip list | grep web3\``. If not found, install it:

```
```bash
pip install web3
```
```

3

```
Create a Python script (main.py)
```

Create a main.py file with the following code:

```
```python
from web3 import Web3
```

```
w3 = Web3(Web3.HTTPProvider("YOUR_QUICKNODE_ENDPOINT_URL/"))
```

```
try:
```

```
    latest_block_number = w3.eth.block_number
```

```
    print(f"Latest Block Number: {latest_block_number}")
```

```
    print(f"Block number (hex): {hex(latest_block_number)}")
```

```
except Exception as e:
```

```
    print(f"Error: {e}")
```
```

```
Eth.rb
```

1

```
Set up your project
```

Create a new directory for your eth.rb project:

```
```bash
mkdir ethereum-ethrb-quickstart

cd ethereum-ethrb-quickstart
```
```

2

### Install eth.rb (if not already installed)

Check if eth.rb is installed with `\`gem list eth\``. If not found, install it:

```
```bash
gem install eth
```
```

3

### Create a Ruby script (main.rb)

Create a main.rb file with the following code:

```
```javascript
require 'eth'
```

```
client = Eth::Client.create "YOUR_QUICKNODE_ENDPOINT_URL/"
```

```
begin
```

```
  block_number = client.eth_block_number
```

```
  hex_result = block_number["result"]
```

```
  decimal_result = hex_result.to_i(16)
```

```
  puts "Block number (hex): #{hex_result}"
```

```
  puts "Block number (decimal): #{decimal_result}"
```

```
rescue => e
```

```
  puts "Error: #{e.message}"
```

```
end
```

```
```
```

4

### Run the script

Execute your Ruby script to see the latest block number:

```
```bash
ruby main.rb
```
```

**\*\*Ethgo\*\***

1

### Set up your project

Create a new directory for your Ethgo project:

```
```bash
mkdir ethereum-ethgo-quickstart

cd ethereum-ethgo-quickstart
```
```

2

### Initialize Go module and install ethgo

Initialize a go.mod file to track dependencies, then install the ethgo SDK:

```
```bash
go mod init ethereum-ethgo-quickstart

go get github.com/umbracle/ethgo

go mod tidy
```
```

3

### Create a Go script (main.go)

Create a main.go file with the following code:

```
```go
package main

import (
    "fmt"

    "log"

    "github.com/umbracle/ethgo/jsonrpc"
)

func main() {

    client, err := jsonrpc.NewClient("YOUR_QUICKNODE_ENDPOINT_URL/")

    if err != nil {

        log.Fatalf("Failed to connect to Ethereum: %v", err)

    }

    blockNumber, err := client.Eth().BlockNumber()
```

```

if err != nil {
    log.Fatalf("Failed to get latest block number: %v", err)
}

fmt.Printf("Latest block number (decimal): %d\n", blockNumber)
fmt.Printf("Latest block number (hex): 0x%x\n", blockNumber)
}

```

4

Run the script

Execute your Go script to see the latest block number:

```

```bash
go run main.go
```

```

If you want to continue learning about making API requests, check out our [guides](<https://quicknode.com/guides>) and [sample apps](<https://www.quicknode.com/sample-app-library/welcome>).

- [API Overview](/docs/ethereum/api-overview): Discover Supported Ethereum APIs, Networks, Chain Configurations, and compatible Quicknode Products
- [Provider Endpoints](/docs/ethereum/endpoints): Learn how to set up and configure your Ethereum RPC endpoints for development and production use
- [Endpoint Security](/docs/ethereum/endpoint-security): Secure your endpoints with Token Authentication, Rate-limiting, Whitelisting, JWT and best security practices
- [Error Reference](/docs/ethereum/error-references): Reference document for common Ethereum RPC error codes and troubleshooting tips
- [Add-ons](/docs/ethereum/add-ons): Discover endpoint add-ons that extend your Ethereum development workflow
- [Guides](/guides/tags/ethereum): Explore comprehensive Ethereum development guides and best practices to accelerate your blockchain application development